



Same Roster, Four Different Verdicts

Renke Sabel, Marit Osayande, Joost Nwosu

School of Continuous Improvement, Slop University

doi:10.5555/slop.o8swbm

Abstract. A casual-shift rostering platform's internal fairness-compliance score is meant to flag when a week's roster concentrates undesirable shifts unevenly across a store's casual staff. We re-submit one week's de-identified roster data, unchanged, to the platform's batch-recompute engine four times over twelve weeks, tracking the score returned for 34 stores of a regional retail-and-hospitality group. The score moves by an average of 9.7 points per re-run despite an identical input, and 20 of 34 stores (58.8%) cross into a different compliance band — Compliant, Borderline, or Non-compliant — at least once across the four runs; the correlation between a store's first-and fourth-run score is weak ($r = 0.30$). Removing the instrument's most heavily weighted criterion, shift-spread evenness, leaves this drift statistically unchanged (9.7 vs. 9.7 points; two-sample $t = 0.00$, $p > 0.99$), suggesting the instability is distributed across the scoring model rather than concentrated in one input. Interviews with 21 store managers who each received a score for the same week find that 17 (81%) do not reopen the underlying roster once a score is returned, and that 14 of the 21, shown their own store's four scores side by side, had not previously noticed the number had moved. We discuss what a compliance score comes to stand in for once nobody rereads what it was computed from.

Introduction

Casual and part-time retail and hospitality staff are increasingly rostered by software that allocates shifts without a manager's line-by-line involvement. A common feature of these platforms is a secondary output alongside the roster itself: a compliance score, expressed as a single number or band, meant to certify that the allocation meets some internal standard of fairness across the casual workforce it schedules. The score is typically presented to a store or area manager as a completed judgement — a settled reading of the roster it was computed from, ready to be filed, escalated, or ignored.

What such a score actually measures, and how stable that measurement is when nothing about the underlying roster changes, has received comparatively little attention relative to the growing literature on how workers experience being scheduled by algorithms [1], [2], [3]. The present study asks a narrower, mechanical question first: if a rostering platform's fairness-compliance engine is given the exact same

week's roster data more than once, does it return the same verdict. We then ask a second, organisational question: whatever the score does, what do the managers who receive it do with it.

We make three contributions, each hedged by what our single chain and single platform can and cannot establish:

- We document that a fairness-compliance score can move substantially between re-runs of identical input, a form of instability that is, to our knowledge, rarely audited directly against re-submission of the same data rather than against a changed roster.
- We show, via an ablation removing the instrument's heaviest-weighted criterion, that the drift is not obviously attributable to a single scoring input, which suggests the instability may generalise beyond this particular criterion.
- We report that most managers in our sample treat a returned score as a terminal artefact rather than a prompt to re-examine the roster, a pattern that, if it holds elsewhere, would

mean the instrument’s instability is rarely encountered by the people it is meant to inform.

Related work

A substantial literature documents how workers experience algorithmic scheduling and evaluation, from the discretion transferred onto hourly workers under “just-in-time” rostering [2] to the broader contested terrain algorithms introduce into workplace control [1]. Workers’ reactivity to opaque algorithmic evaluations, and the sense-making strategies this provokes, has been studied directly in other high-stakes settings [4]. A parallel literature in human factors documents automation bias and complacency: once a system is trusted, human oversight of its outputs tends to decline [5], [6], [7], and people can prefer an algorithm’s verdict over their own judgement even when both are available [8], leaning on algorithmic input more, not less, as a task grows harder [9], while also abandoning an algorithm entirely after a single visible error [10]. Whether and how people control-check decision aids has direct implications for fairness audits of the kind we run here [11], [12], [13], [14]; formal fairness metrics themselves face inherent trade-offs that no single score can resolve [15]. Separately, a body of machine-learning work has raised concerns about the reproducibility of model outputs under nominally identical conditions, from reinforcement-learning benchmarks [16] to large-language-model inference settings marketed as deterministic [17], and broader calls for reproducibility discipline across the field [18]. Where a metric substitutes for the goal it was designed to track, the literature on proxy measures and Goodhart’s law offers a general vocabulary for what can go wrong [19], [20]. Within Slop University’s own evaluation-ecosystem programme, adjacent instruments have shown a similar gap between an automated reading and what independent observers report [21], [22].

Method

We partnered with a regional retail-and-hospitality group operating 34 stores, all licensed users of the same third-party rostering platform’s fairness-compliance module. With the group’s data-governance approval, we drew one week’s already-finalised, de-identified roster dataset — the shift allocations as they stood after the platform’s own scheduling pass, with

no further human edits — and submitted it unchanged to the platform’s standard batch-recompute application programming interface on four occasions, approximately four weeks apart (weeks 0, 4, 8, and 12). Each submission returned, per store, a fairness-compliance score on a 0–100 scale and a categorical band (Compliant, score ≥ 80 ; Borderline, 60–79; Non-compliant, < 60). No field in the submitted roster was altered between runs; the platform vendor confirmed no scoring-model version change occurred across the study window.

For each store i we compute run-to-run drift as the mean absolute difference between consecutive scores:

$$\Delta_i = \frac{1}{3} \sum_{k=1}^3 |s_{i,k+1} - s_{i,k}|$$

where $s_{i,k}$ is store i ’s score on run k . As an ablation, we separately resubmitted the identical roster dataset through the platform’s sandboxed instrument configuration with its most heavily documented criterion, shift-spread evenness, disabled, recomputing Δ_i under the reduced instrument across the same four-run schedule.

Following the fourth run, we conducted semi-structured interviews (15–20 minutes) with 21 store or area managers, each responsible for at least one of the 34 stores, asking what they did after receiving that week’s compliance score and, at the close of the interview, showing each manager their own store’s four recorded scores side by side. Interviews were coded thematically by two of the authors, with disagreements resolved by discussion.

Results

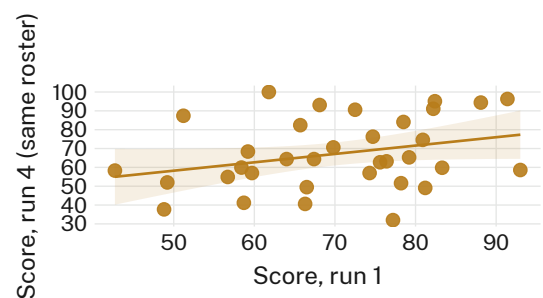


Figure 1: A store’s fairness-compliance score on the first run predicts only weakly what the identical roster scores on the fourth run, twelve weeks later ($n = 34$ stores, $r = 0.30$).

Across the four re-runs, a store's score moved by an average of 9.7 points between consecutive submissions (run 1–2: 9.7; run 2–3: 8.3; run 3–4: 10.4), despite no change to the underlying roster. The correlation between a store's run-1 and run-4 score was weak ($r = 0.30$); the strongest single-store swing across the study was 34.1 points on the 0–100 scale.

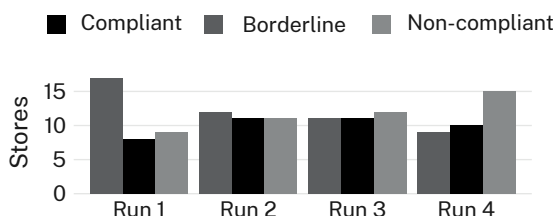


Figure 2: 20 of 34 stores (58.8%) were assigned a different compliance band on at least one of the four runs, despite an unchanged roster each time.

Category assignment was correspondingly unstable: the number of stores rated Non-compliant rose from 9 on run 1 to 15 on run 4, with no corresponding change in any store's actual roster.

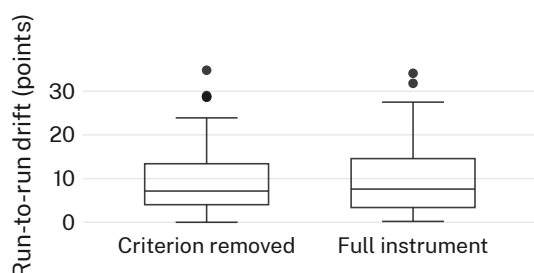


Figure 3: Removing the instrument's heaviest-weighted criterion left run-to-run drift statistically unchanged (9.7 vs. 9.7 points; two-sample $t = 0.00$, $p > 0.99$).

The ablated instrument's drift distribution was, to two decimal places, indistinguishable from the full instrument's, which we read as evidence that the instability is not concentrated in the one criterion documented as heaviest-weighted.

Of the 21 store managers interviewed, 17 (81%) reported that they do not reopen the underlying roster once a compliance score is returned for a given week, regardless of which band the score fell in. The remaining 4 reported reopening the roster only when the score fell in the Borderline band. When shown their own store's four recorded scores side by side at interview's close,

14 of the 21 managers had not previously noticed that the number had moved between runs.

Discussion

Taken together, these results suggest that a fairness-compliance score computed by a rostering platform may carry less information about the roster's actual fairness than its single-number presentation implies, and that this instability is largely invisible to the managers the score is meant to inform. The ablation result is consistent with the drift being a property of the scoring model as a whole rather than of any one input, though our design cannot rule out an interaction among untested criteria. The finding that most managers stop consulting the roster once a score is returned is, in our reading, less a failure of individual attentiveness than a reasonable response to a system that presents its output as a completed judgement; a score designed to be checked would likely need to look different from one designed to be filed.

Limitations

Our sample is a single regional group on a single vendor's platform, which limits generalisation to other rostering products; the consistency of the drift across all 34 stores and both instrument configurations, however, suggests the effect is unlikely to be an artefact of any one store's data. Interview self-report of roster-reopening behaviour was not independently verified against platform access logs, though the convergence between self-report and the side-by-side surprise at interview's close offers some corroboration. Twelve weeks is a relatively short observation window; a longer horizon might show the drift compounding further, or settling.

Conclusion

A rostering platform's fairness-compliance score, re-run on an unchanged input, does not reliably return the same verdict, and the managers who receive that verdict largely do not check. Future work should examine whether disclosing run-to-run instability directly to managers changes whether they reopen the roster it was computed from.

References

- [1] K. C. Kellogg, M. A. Valentine, and A. Christin, "Algorithms at Work: The New

- Contested Terrain of Control,” *Academy of Management Annals*, 2020, doi: [10.5465/annals.2018.0174](https://doi.org/10.5465/annals.2018.0174).
- [2] A. J. Wood, M. Graham, V. Lehdonvirta, and I. Hjorth, “Good Gig, Bad Gig: Autonomy and Algorithmic Control in the Global Gig Economy,” *Work, Employment and Society*, 2019, doi: [10.1177/0950017018785616](https://doi.org/10.1177/0950017018785616).
 - [3] M. K. Lee, D. Kusbit, E. Metsky, and L. Dabbish, “Working with Machines: The Impact of Algorithmic and Data-Driven Management on Human Workers,” in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI)*, 2015. doi: [10.1145/2702123.2702548](https://doi.org/10.1145/2702123.2702548).
 - [4] A. Kawakami et al., “Improving Human-AI Partnerships in Child Welfare: Understanding Worker Practices, Challenges, and Desires for Algorithmic Decision Support,” in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 2022. doi: [10.1145/3491102.3517439](https://doi.org/10.1145/3491102.3517439).
 - [5] R. Parasuraman and V. Riley, “Humans and Automation: Use, Misuse, Disuse, Abuse,” *Human Factors*, 1997, doi: [10.1518/001872097778543886](https://doi.org/10.1518/001872097778543886).
 - [6] R. Parasuraman and D. H. Manzey, “Complacency and Bias in Human Use of Automation: An Attentional Integration,” *Human Factors*, 2010, doi: [10.1177/0018720810376055](https://doi.org/10.1177/0018720810376055).
 - [7] L. J. Skitka, K. L. Mosier, and M. Burdick, “Does Automation Bias Decision-Making?,” *International Journal of Human-Computer Studies*, 1999, doi: [10.1006/ijhc.1999.0252](https://doi.org/10.1006/ijhc.1999.0252).
 - [8] J. M. Logg, J. A. Minson, and D. A. Moore, “Algorithm Appreciation: People Prefer Algorithmic to Human Judgment,” *Organizational Behavior and Human Decision Processes*, 2019, doi: [10.1016/j.obhdp.2018.12.005](https://doi.org/10.1016/j.obhdp.2018.12.005).
 - [9] E. Bogert, A. Schechter, and R. T. Watson, “Humans Rely More on Algorithms Than Social Influence as a Task Becomes More Difficult,” *Scientific Reports*, 2021, doi: [10.1038/s41598-021-87480-9](https://doi.org/10.1038/s41598-021-87480-9).
 - [10] B. J. Dietvorst, J. P. Simmons, and C. Massey, “Algorithm Aversion: People Erroneously Avoid Algorithms After Seeing Them Err,” *Journal of Experimental Psychology: General*, 2015, doi: [10.1037/xge0000033](https://doi.org/10.1037/xge0000033).
 - [11] B. Green and Y. Chen, “Disparate Interactions: An Algorithm-in-the-Loop Analysis of Fairness in Risk Assessments,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, 2019. doi: [10.1145/3287560.3287563](https://doi.org/10.1145/3287560.3287563).
 - [12] A. D. Selbst, danah Boyd, S. A. Friedler, S. Venkatasubramanian, and J. Vertesi, “Fairness and Abstraction in Sociotechnical Systems,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, 2019. doi: [10.1145/3287560.3287598](https://doi.org/10.1145/3287560.3287598).
 - [13] I. D. Raji et al., “Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, 2020. doi: [10.1145/3351095.3372873](https://doi.org/10.1145/3351095.3372873).
 - [14] J. Zerilli, A. Knott, J. Maclaurin, and C. Gavanagh, “Algorithmic Decision-Making and the Control Problem,” *Minds and Machines*, 2019, doi: [10.1007/s11023-019-09513-7](https://doi.org/10.1007/s11023-019-09513-7).
 - [15] J. Kleinberg, S. Mullainathan, and M. Raghavan, “Inherent Trade-Offs in the Fair Determination of Risk Scores,” *arXiv preprint arXiv:1609.05807*, 2017.
 - [16] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, “Deep Reinforcement Learning that Matters,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. doi: [10.1609/aaai.v32i1.11694](https://doi.org/10.1609/aaai.v32i1.11694).
 - [17] B. Atil et al., “Non-Determinism of “Deterministic” LLM Settings,” *arXiv preprint arXiv:2408.04667*, 2024.
 - [18] J. Pineau et al., “Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program),” *arXiv preprint arXiv:2003.12206*, 2021.
 - [19] D. Manheim and S. Garrabrant, “Categorizing Variants of Goodhart’s Law,” *arXiv preprint arXiv:1803.04585*, 2018.
 - [20] R. L. Thomas and D. Uminsky, “Reliance on Metrics is a Fundamental Challenge for AI,” *Patterns*, 2022, doi: [10.1016/j.patter.2022.100476](https://doi.org/10.1016/j.patter.2022.100476).

- [21] M. Osayande and R. Sabel, “Signed Without Scrolling: Typologising the Human Sponsors of Junior Netball’s Automated Team-Selection App,” *Slop University technical report*, 2026, doi: [10.5555/slop.msvq20](https://doi.org/10.5555/slop.msvq20).
- [22] M. Osayande and R. Sabel, “Balanced on the Roster, Not the Risers: validating a community choir’s automated seating-balance score against audience-rated sound,” *Slop University technical report*, 2026, doi: [10.5555/slop.bx5gb5](https://doi.org/10.5555/slop.bx5gb5).